Workflow Management in Open OnDemand

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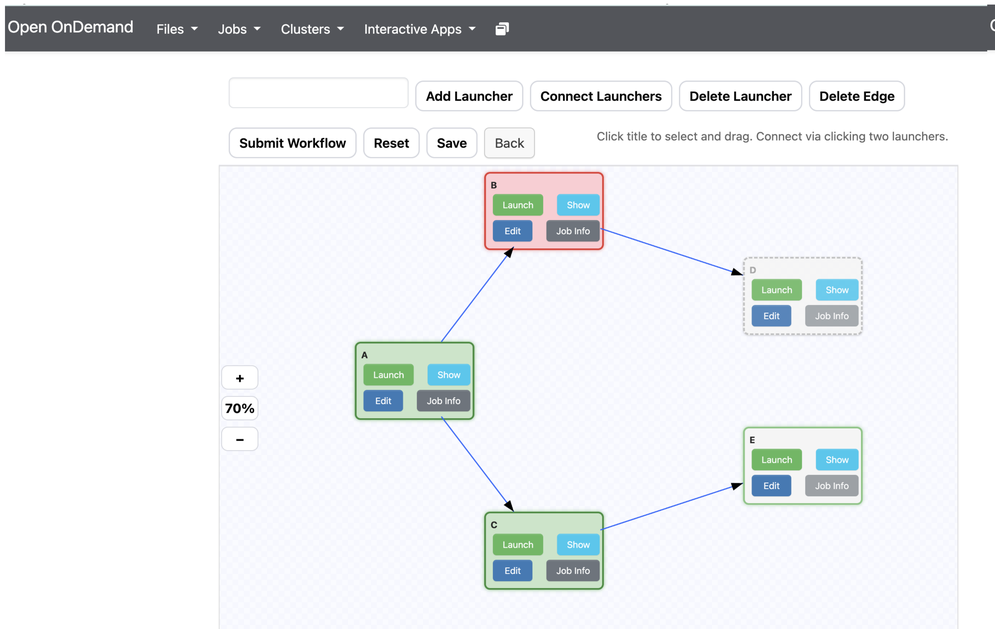
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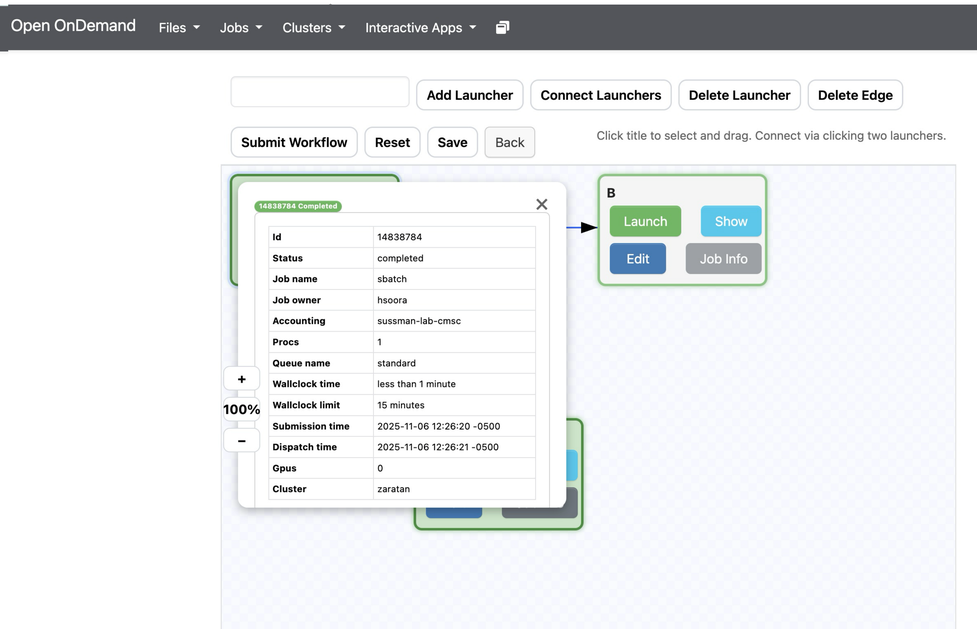
Overview

The *Workflow Manager* enhancement for Open OnDemand extends the Project Manager feature to support complex, multi-stage computational workflows, enabling researchers to visually design, configure, and execute interconnected tasks through a web-based interface.

Open OnDemand is used by over 700 sites in 62 countries, faces growing demands from researchers for intuitive, flexible, and reproducible workflow orchestration.

Users can chain together multiple tasks, with conditional branching based on job success or failure. Each launcher provides actions such as Launch, View, Edit, and Job Info, allowing fine-grained control for each task while maintaining dependencies between tasks and overall workflow execution.



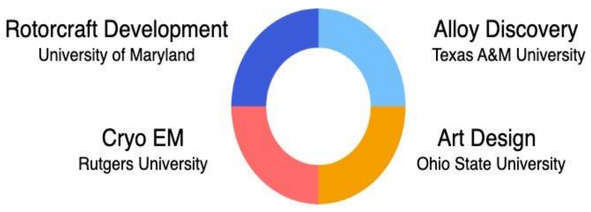


OBJECTIVE

Users should be able to:

- Construct flexible, multi-stage workflows using both existing and new job launchers.
- Define dependencies among tasks including conditional branches for success and failure handling of individual tasks.
- Store, share, and reproduce complex research pipelines as reusable workflows.
- Monitor job status and progress in real-time, with clear reporting of errors and availability of intermediate outputs.

Scientific Application Collaborators

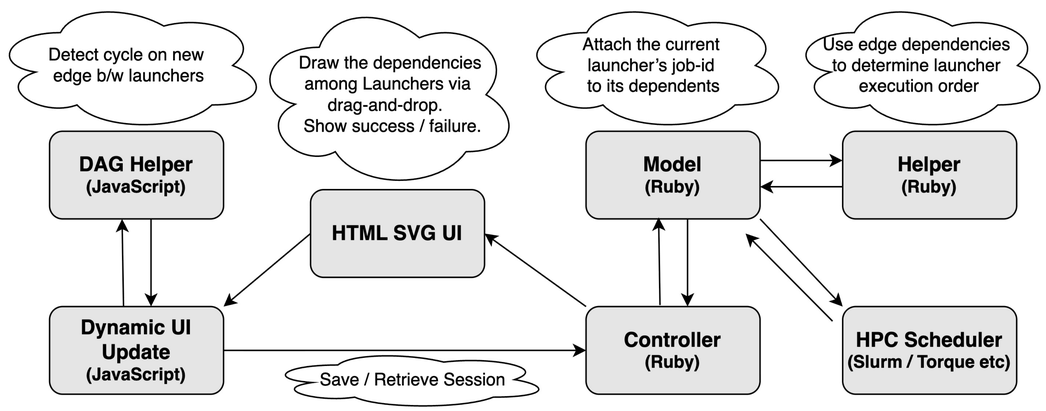


DESIGN & IMPLEMENTATION

- **Workflow MVC:** A dedicated workflow class to store a subset of launchers and their relationships as a directed acyclic graph of nodes (tasks) and edges (dependencies).

  YAML configuration files store workflow metadata, supporting easy serialization, editing, reproducibility.

- **Job Status Reporting:** Tasks within a workflow are captured and status exposed to users in real time.
- **Job Submission Logic:** Existing OOD job scheduling adaptor ensures broad compatibility with HPC clusters and resource managers. Eg. Slurm, Torque, PBS Pro.
- **UI Integration:** Collapsible lists, intuitive icons, and context help minimize interface clutter.
- **Job Status Reporting:** Tasks within a workflow are captured and status exposed to the user in real time.



Design Diagram

FUTURE WORK

- A preview of Workflow Manager is now available on the master branch of OnDemand GitHub repository.
- It will be released for community feedback in v4.1
- Extend workflow editing features to support merging and cloning existing workflows, enabling reuse and ensemble simulations with shared or variant launchers

REFERENCES

- https://github.com/OSC/ondemand/issues/4338
- https://openondemand.org/

GitHub

ACKNOWLEDGEMENT

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It calls for "cross-cutting efforts" to deliver customizable toolsets for science drivers across disciplines.

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